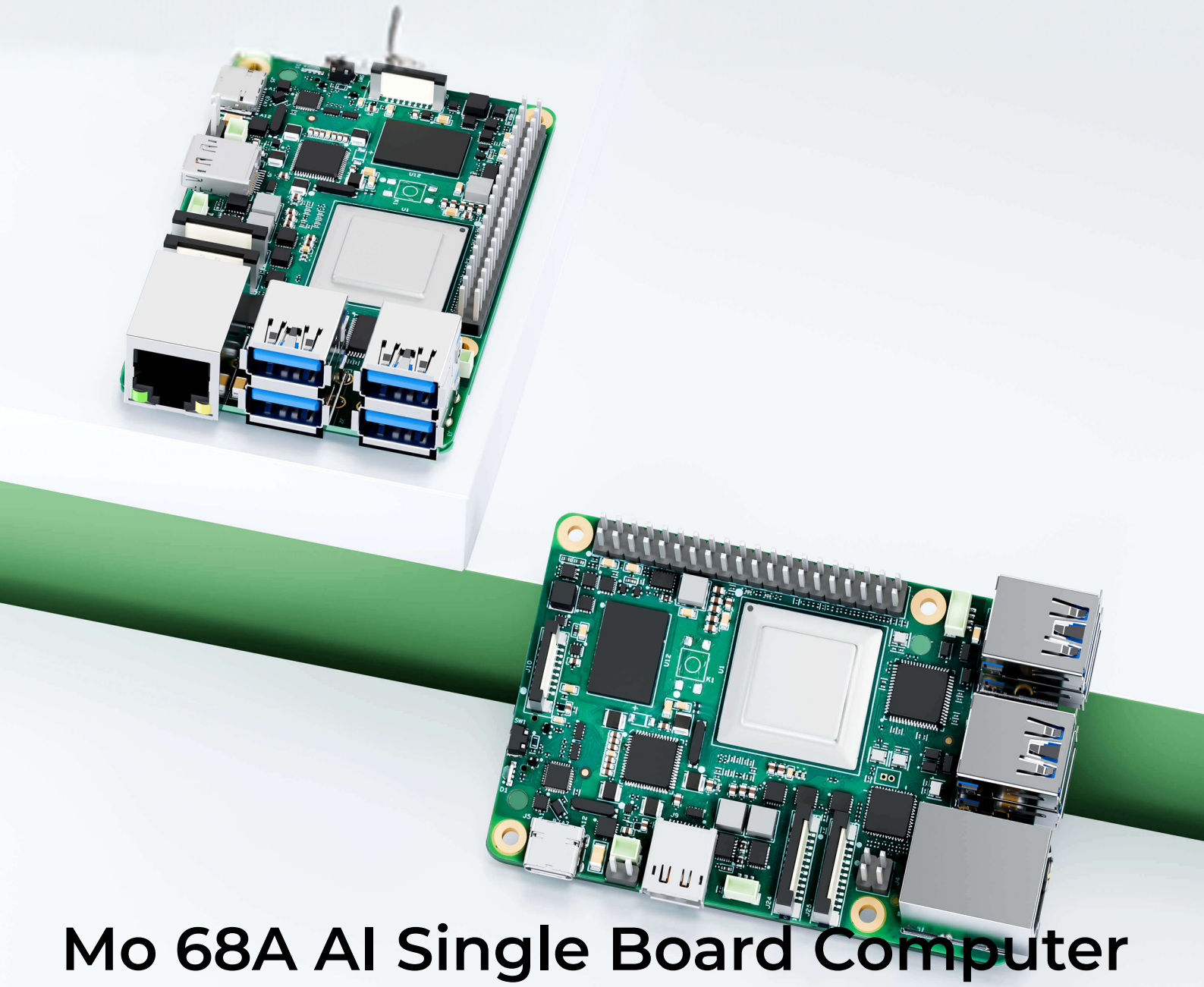
InHand Networks

Embrace Edge AI, Empower Intelligent Vision



Mo 68A AI Single Board Computer

· Open SDK

· Linux Distribution

· HAT Expansion

· Edge AI

1. Product Overview

The Mo 68A is an 8 TOPS edge AI single-board computer based on the TI AM68A SoC, with a form factor compatible with standard SBCs, designed for AI vision boxes, intelligent cameras, and edge AI inference terminals.

Product Features:

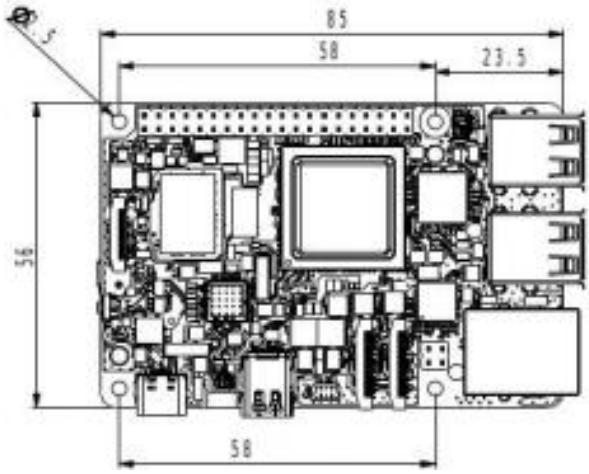
- **Edge AI:** 8 TOPS on-device AI acceleration, TI TIDL supports TFLite / ONNX
- **Vision Processing:** Dual 4-lane MIPI CSI-2, VPAC / DMPAC, 4K@60fps codec
- **Rich Interfaces:** Gigabit Ethernet, PCIe 3.0, 4 × USB 3.0, mini DP
- **SBC Compatibility:** Standard form factor, 40-pin HAT compatibility, support for mainstream enclosures and accessories
- **High Security:** Secure Boot, TrustZone, OP-TEE, hardware crypto acceleration

Core Technical Specifications

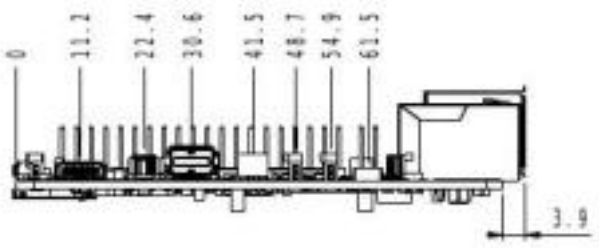
Technical Indicator	Specification
AI Inference	8 TOPS, TI TIDL supports TFLite / ONNX models
Vision Processing	Dual 4-lane MIPI CSI-2 RX, VPAC / DMPAC, 4K@60fps H.265 / H.264 hardware codec
Operating System	Debian 13 Trixie, Linux 6.12
Network	1 × Gigabit Ethernet
Security	Secure Boot, TrustZone, OP-TEE, AES-256
Development	Python / C / C++, OpenCV, Open SDK
CPU	2 × Cortex-A72 @ 2.0 GHz
Memory	LPDDR4 4 GB / 8 GB (default)
Storage	Micro SD

Technical Indicator	Specification
Power	USB Type-C 5 V / 5 A DC, 25 W (MAX)
Dimensions	85 × 56 mm
Operating Temperature	0 ~ 50 °C

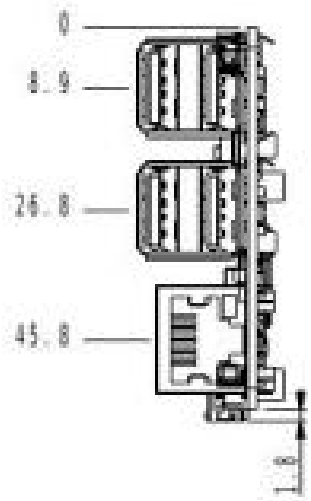
2. Product Dimensions



Front View



Side View



Interface View

Note:

1. All dimensions are in millimeters (mm).
2. All dimensions are approximate values for **reference only**.
3. The dimensions shown **shall not be used for production or processing**.
4. Dimensions shall comply with part and manufacturing tolerance requirements.
5. Dimensions are subject to change without notice.

3. Hardware Specifications

Category / Parameter	Specification
Hardware Platform	
CPU	TI AM68A, 2 × Cortex-A72 @ 2.0 GHz
AI Accelerator	2 × C7x DSP + Deep Learning Accelerator, 8 TOPS
ISP / Vision	On-chip ISP + VPAC (RGB-IR, WDR, LDC)
RAM	LPDDR4 4 GB / 8 GB (default)
Interfaces	
Ethernet	1 × Gigabit Ethernet
USB	4 × USB 3.0 Type-A
PCIe	1 × PCIe 3.0
Display	1 × mini DP, up to 2 × 4-lane MIPI DSI
Camera	up to 2 × 4-lane MIPI CSI-2
Audio	I2S via 40-pin connector
40-pin Connector	GPIO / I2C / I2S / SPI / UART / PCM, HAT-compatible
Fan Connector	1 × 4-pin fan connector (5 V, PWM, GND, TACH)
Button	1 × Reset button
Storage	Micro SD
Debug UART	1 × TTL UART
LED	PWR, STATUS
Security	

Category / Parameter	Specification
Secure Boot	Support (ARM TrustZone + OP-TEE)
Power	
Power Input	USB Type-C 5 V / 5 A DC
Power Consumption	25 W (MAX)
Mechanical	
Dimensions	85 × 56 mm
Weight	53 g
Housing	PCB
Cooling	Active fan (optional)
Others	
RTC	Support (battery backup)
Watchdog	Hardware watchdog supported
Environmental	
Operating Temperature	0 ~ 50 °C
Storage Temperature	-20 ~ 70 °C

4. Software Specifications

Category / Parameter	Specification
Operating System	
OS	Debian 13 Trixie

Category / Parameter	Specification
Kernel	Linux Kernel 6.12
AI & Vision	
AI Runtime	TI TIDL, supports TFLite / ONNX
Vision SDK	TI EdgeAI SDK
Camera Framework	V4L2
Display Framework	DRM / KMS
Network Features	
IP Application	TCP / UDP, ICMP, DNS, DHCP
IP Routing	Static routing
Security	
Secure Boot	Support
TrustZone	Support
OP-TEE	Support
Crypto Accelerator	Hardware AES-256
Development	
Languages	Python, C / C++
Libraries	OpenCV, GStreamer, NumPy
Package Manager	apt (Debian)
Open SDK	Supports custom system build by customer
System Management	
Remote Access	SSH

Category / Parameter	Specification
Firmware Upgrade	SD card flash
Debug	UART console

5. Ordering Information

Model Code Rule

Model code: Mo68A-<x>-<y>

Mo68A: Base Model

<-x>: Memory

<-y>: MIPI configuration

Product Models

Base Model	-x	Memory	-y	MIPI Configuration
Mo68A	4G	4 GB	DC	DSI on MIPI0, CSI on MIPI1
Mo68A	8G	8 GB	DC	DSI on MIPI0, CSI on MIPI1

Example
Model: Mo68A-8G-DC
"Mo68A" means Base Model.
"-8G" means 8 GB memory.
"-DC" means DSI on MIPI0, CSI on MIPI1.

6. Contact Us

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